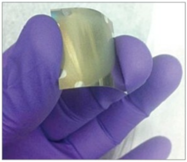
Lab-grown LED provides better way to clean water



Ohio State University researchers have developed a technique to create LEDs on metal foil (Image courtesy: Ohio State University)

Engineers at Ohio State University, USA, have developed foil based LEDs for portable ultraviolet (UV) lights that soldiers and others can use to purify drinking water and sterilise medical equipment. The LEDs are created on lightweight flexible metal foil.

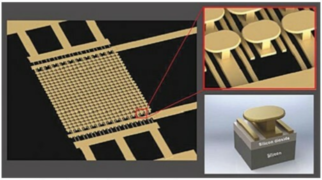
As per Roberto Myers, associate professor of materials science and engineering at Ohio State University, the military, industry and humanitarian organisations already use deep UV light for applications ranging from detection of biological agents to curing plastics. However, traditional deep-UV lamps are very heavy and electrically inefficient.

LED foil could offer a more environmentally-friendly light to purify water. The researchers are confident that they will be able to scale up their prototype; their goal is to transform nanophotonics, a study centred around how objects just nanometres big interact with light, into a profitable industry.

World's first semiconductor-free microelectronic device is here

Engineers at University of California, San Diego, USA, have created the first semiconductor-free, optically-controlled microelectronic device. Using metamaterials, they were able to build a micro-scale device that shows a 1000 per cent increase in conductivity when activated by low voltage and a low power laser.

Capabilities of existing microelectronic devices such as transistors are ultimately limited by the properties of their constituent materials such as their semiconductors, according to the researchers.



Semiconductor-free microelectronic device (Image courtesy: UC San Diego Applied Electromagnetics Group)

A team of researchers in Applied Electromagnetics Group led by electrical engineering professor Dan Sievenpiper at UC San Diego, in order to remove these roadblocks to conductivity, replaced semiconductors with free electrons in space.

The discovery paves the way for microelectronic devices that are faster and capable of handling more power, and could also lead to more efficient solar panels.

A drone that flies like a bird

A drone developed by Dario Floreano's team in Laboratory of Intelligent Systems at École Polytechnique Fédérale de Lausanne, Switzerland, is geared toward flying more like a bird than like small helicopters, as most available multi-rotors currently do.

After observing birds in flight, researchers from Laboratory of Intelligent Systems had the idea of building an energy-efficient winged drone capable of changing its wingspan, flying at high speed and moving through tight spaces (Image courtesy: École Polytechnique Fédérale de Lausanne)

Floreano and his team wanted to develop a bio-inspired drone that could meet various aerodynamic requirements. It had to be capable of flying between obstacles, making sharp turns and coping with strong winds. By changing its geometry mid-flight, their drone can meet all these criteria. The moving part is located on the outer wings. It works like a bird's quill feathers, which are the large feathers at the edge of the wing.

Ford using drones to guide self-driving cars

Automobile giant Ford is studying a system through which it can use drones to help self-driving cars. The system also guides the vehicles in off-road adventures.

Drones launched from an autonomous vehicle would help guide it by mapping the surrounding area beyond what the car's sensors can detect. Passengers can control the drone using the car's infotainment or navigation system.

According to a company spokesperson, the drones could also prove useful in areas beyond the digital maps of urban and sub-urban areas and inter-city highways.